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Section: C
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① ~~Prove that f(z)~~

Given, $f(z) = |z|^2$.

According to Laplace equation –

$$\frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2} = 0$$

and,

$$\frac{\delta^2 v}{\delta x^2} + \frac{\delta^2 v}{\delta y^2} = 0$$

$$|z| = \sqrt{x^2 + y^2} \quad \text{or} \quad |z|^2 = x^2 + y^2.$$

Let's assume, $f(x, y) = x^2 + y^2$

Then,

$$\frac{\delta f}{\delta x} = \frac{\delta}{\delta x}(x^2 + y^2) = 2x, \quad \text{or} \quad \frac{\delta^2 f}{\delta x^2} = 2$$

$$\frac{\delta f}{\delta y} = \frac{\delta}{\delta y}(x^2 + y^2) = 2y \quad \text{or} \quad \frac{\delta^2 f}{\delta y^2} = 2$$

$2 + 2 = 4 \neq 0$ ∴ not harmonic.

$$f(z) = \ln(|z|^2)$$

$$f = \ln(x^2+y^2)$$

or, $f(x,y) = \ln(x^2+y^2)$

$$\frac{\delta f}{\delta x} = \frac{1}{x^2+y^2} \cdot 2x = \frac{2x}{x^2+y^2}.$$

$$\frac{\delta f}{\delta y} = \frac{1}{x^2+y^2} \cdot 2y = \frac{2y}{x^2+y^2}.$$

$$= \frac{2x}{(x^2+y^2)^2} \frac{\delta}{\delta x}(x^2+y^2)$$

$$\frac{\delta^2 f}{\delta x^2} = \frac{\delta}{\delta x}\left(2x \cdot \frac{1}{x^2+y^2}\right) = \frac{2}{x^2+y^2} + \left(-\frac{4x^2}{(x^2+y^2)^2}\right)$$

$$= \frac{2}{x^2+y^2} - \frac{4x^2}{(x^2+y^2)^2}$$

$$= \frac{2(x^2+y^2) - 4x^2}{x^2+y^2}$$

$$= \frac{2y^2-2x^2}{x^2+y^2}.$$

Similarly, $\frac{\delta^2 f}{\delta y^2} = \frac{2x^2-2y^2}{x^2+y^2}$

$$\frac{\delta^2 f}{\delta x^2} + \frac{\delta^2 f}{\delta y^2} = \frac{2y^2-2x^2}{x^2+y^2} + \frac{2x^2-2y^2}{x^2+y^2} = 0$$

$\therefore$ Harmonic

(ii) Given, $f(x,y,z) = x^2+y^2-2z^2$.

$$\frac{\delta f}{\delta x} = 2x \; ; \; \frac{\delta f}{\delta y} = 2y \; ; \; \frac{\delta f}{\delta z} = -4z$$

$$\frac{\delta^2 f}{\delta x^2} = 2 \; ; \; \frac{\delta^2 f}{\delta y^2} = 2 \; ; \; \frac{\delta^2 f}{\delta z^2} = -4$$

Here, $\frac{\delta^2 f}{\delta x^2} + \frac{\delta^2 f}{\delta y^2} + \frac{\delta^2 f}{\delta z^2} = 2+2-4 = 0$

$\therefore$ Harmonic.

$f(x,y,z) = x^2 - y^2 + z^2$.

$$\frac{\delta f}{\delta x} = 2x \; ; \; \frac{\delta f}{\delta y} = -2y \; ; \; \frac{\delta f}{\delta z} = 2z$$

$$\frac{\delta^2 f}{\delta x^2} = 2 \; ; \; \frac{\delta^2 f}{\delta y^2} = -2 \; ; \; \frac{\delta^2 f}{\delta z^2} = 2.$$

Here, $\frac{\delta^2 f}{\delta x^2} + \frac{\delta^2 f}{\delta y^2} + \frac{\delta^2 f}{\delta z^2} = 2-2+2 = 2$

$\therefore$ Not harmonic.

(iii) $u(x,y) = 3x^3 - 9xy^2$

Here,

$$\frac{\delta u}{\delta x} = 9x^2 - 18y\ 9y^2$$

$$\frac{\delta^2 u}{\delta x^2} = \overset{18x}{\cancel{8}} - 9y^2.$$

$$\frac{\delta u}{\delta y} = 3x^3 - 18xy$$

$$\frac{\delta^2 \cancel{9}u}{\delta y^2} = 3x^3 - 18x.$$

(iii) $u(x,y) = 3x^3 - 9xy^2$

Here, $$\frac{\delta u}{\delta x} = 9x^2 - 9y^2.$$

$$\frac{\delta^2 u}{\delta x} = 18x.$$

$$\frac{\delta u}{\delta y} = -18xy.$$

$$\frac{\delta^2 u}{\delta y^2} = -18x.$$

$$\frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2} = 18x - 18x = 0$$

$\therefore$ Harmonic.

(iv) $u(x,y) = x^3 - 3xy^2 - 5y.$

$$\frac{\delta u}{\delta x} = 3x^2 - 3y^2$$

$$\frac{\delta^2 u}{\delta x^2} = 6x.$$

$$\frac{\delta u}{\delta y} = -6xy - 5$$

$$\frac{\delta^2 u}{\delta y^2} = -6x$$

$$\frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2} = 6x - 6x = 0$$

$\therefore$ Harmonic.

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(v) (a) $u = x^3 - 3xy^2$

$$\frac{\delta u}{\delta x} = 3x^2 - 3y^2 \rightarrow \frac{\delta^2 u}{\delta x^2} = 9x.$$

$$\frac{\delta u}{\delta y} = -6xy \rightarrow \frac{\delta^2 u}{\delta y^2} = -6x.$$

$$\frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2} = 9x - 6x = 3x$$

$\therefore$ Not harmonic.

b) $u = e^{-y} \sin x$.

$$\frac{\delta u}{\delta x} = e^{-y} \cos x.$$

$$\frac{\delta^2 u}{\delta x^2} = -e^{-y} \sin x.$$

$$\frac{\delta u}{\delta y} = -e^{-y} \sin x$$

$$\frac{\delta^2 u}{\delta y^2} = -(-e^{-y} \sin x)$$

$$= e^{-y} \sin x.$$

$$\therefore \frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2}$$

$$= -e^{-y} \sin x + e^{-y} \sin x$$

$$= 0$$

$\therefore$ Harmonic.

(c) $u = e^x \cos y$.

$$\frac{\delta u}{\delta x} = e^x \cos y.$$

$$\Rightarrow \frac{\delta^2 u}{\delta x^2} = e^x \cos y.$$

$$\frac{\delta u}{\delta y} = -e^x \sin y$$

$$\Rightarrow \frac{\delta^2 u}{\delta y^2} = -e^x \cos y.$$

$$\frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2}$$

$$= e^x \cos y - e^x \cos y$$

$$= 0$$

$\therefore$ Harmonic.

(d)

$$u(x,y) = 3x^2y + 2x^2 - y^3 - 2y^2$$

$$\frac{\delta u}{\delta x} = 6xy + 4x$$

$$\Rightarrow \frac{\delta^2 u}{\delta x^2} = 6y + 4.$$

$$\frac{\delta u}{\delta y} = 3x^2 - 3y^2 - 4y$$

$$\Rightarrow \frac{\delta^2 u}{\delta y^2} = -6y - 4.$$

$$\frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2}$$

$$= 6y + 4 - 6y - 4$$

$$= 0$$

$\therefore$ Harmonic.

(e) $u(x,y) = e^x(x\cos y - y\sin y)$

$$\frac{\delta u}{\delta x} = e^x(x\cos y - y\sin y) + e^x\frac{\delta}{\delta x}(x\cos y - y\sin y)$$

$$= u(x,y) = e^x(x\cos y - y\sin y)$$

$$= xe^x\cos y - ye^x\sin y.$$

$$\frac{\delta u}{\delta x} = \frac{\delta}{\delta x}(x)\cdot e^x\cdot\cos y + x\cos y\frac{\delta}{\delta x}(e^x) + xe^x\frac{\delta}{\delta x}(\cos y)$$
$$- \left(e^x\sin y\frac{\delta}{\delta x}(y) + y\sin y\frac{\delta}{\delta x}(e^x) + ye^x\frac{\delta}{\delta x}(\sin y)\right).$$
$$= e^x\cdot\cos y + xe^x\cos y - ye^x\sin y.$$

$$\Rightarrow \frac{\delta^2 u}{\delta x^2} = e^x\cos y + e^x\cos y + xe^x\cos y - ye^x\sin y.$$

← Doin' in short.

$$\frac{\delta u}{\delta y} = -xe^x\sin y - e^x\sin y - ye^x\cos y.$$

One for the Laugh, two for the show

$$\Rightarrow \frac{\delta^2 u}{\delta y^2} = -xe^x\cos y - e^x\cos y + ye^x\sin y - e^x\cos y.$$

$$\frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2}$$

$$= e^x\cos y + e^x\cos y + xe^x\cos y - ye^x\sin y$$
$$- xe^x\cos y - e^x\cos y + ye^x\sin y - e^x\cos y$$

$$= 0$$

$\therefore$ Harmonic

(f) $u = y^3 - 3x^2y$

$$\frac{\delta u}{\delta x} = -6xy$$

$$\frac{\delta^2 x}{\delta x^2} = -6y.$$

$$\frac{\delta u}{\delta y} = 3y^2 - 3x^2$$

$$\frac{\delta^2 u}{\delta y^2} = 6y.$$

$$\therefore \frac{\delta^2 x}{\delta x^2} + \frac{\delta^2 u}{\delta y}$$

$$= -6y + 6x$$

$$= 0$$

$\therefore$ Harmonic.

(g) $u = 2x(1-y)$

$$= 2x - 2xy$$

$$\frac{\delta u}{\delta x} = 2 - 2y$$

$$\Rightarrow \frac{\delta^2 u}{\delta x^2} = 0$$

$$\frac{\delta u}{\delta y} = 0 - 2x$$

$$\frac{\delta^2 u}{\delta y^2} = 0$$

$$\frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2}$$

$$= 0 + 0$$

$$= 0$$

Harmonic.

(h) $u(x,y) = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$

$$\frac{\delta u}{\delta x} = 3x^2 - 3y^2 + 6x$$

$$\Rightarrow \frac{\delta^2 u}{\delta x^2} = 6x + 6.$$

$$\frac{\delta u}{\delta y} = -6xy - 6y$$

$$\Rightarrow \frac{\delta^2 u}{\delta y^2} = -6x - 6$$

$$\frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2}$$

$$= 6x + 6 - 6x - 6$$

$$= 0$$

Harmonic.

(i) $u(x,y)$

$$= x^2 - y^2 - 2xy - 2x + 3y$$

$$\frac{\delta u}{\delta x} = 2x - 2y - 2$$

$$\Rightarrow \frac{\delta^2 u}{\delta x^2} = 2$$

$$\frac{\delta u}{\delta y} = -2y - 2x + 3$$

$$\frac{\delta^2 u}{\delta y^2} = -2$$

$$\frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2}$$

$$= 2 - 2 = 0$$

Harmonic.